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PHYSICS DEPARTMENT

PHY2049, Spring 2021

Final Exam

April 27, 2021

Name (print, last first): _____ Signature: _____

*On my honor, I have neither given nor received unauthorized aid on this examination.***YOUR EXAM NUMBER IS THE 5-DIGIT NUMBER AT THE TOP OF EACH PAGE.****DIRECTIONS**

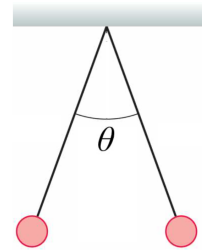
- (1) **Code your exam number on your answer sheet (use 76–80 for the 5-digit number).** Code your name on your answer sheet. **DARKEN CIRCLES COMPLETELY.** Code your student number on your answer sheet.
- (2) Print your name on this sheet and sign it also.
- (3) Do all scratch work anywhere on this exam that you like. At the end of the exam, this exam printout is to be turned in. No credit will be given without both answer sheet and printout with scratch work most questions demand.
- (4) **Blacken the circle of your intended answer completely, using a #2 pencil or blue or black ink.** Do not make any stray marks or the answer sheet may not read properly.
- (5) The answers are rounded off. Choose the closest to exact. There is no penalty for guessing.

>>>>>>>**WHEN YOU FINISH**<<<<<<<
Hand in the exam and answer sheet separately.

Constants		
$m_e = 9.11 \times 10^{-31} \text{ kg}$	$m_p \simeq m_n = 1.67 \times 10^{-27} \text{ kg}$	$e = 1.6 \times 10^{-19} \text{ C}$
$k = 9 \times 10^9 \text{ N m}^2/\text{C}^2$	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$	$\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$
$\mu_B = 9.27 \times 10^{-24} \text{ J/T}$	$c = 3 \times 10^8 \text{ m/s}$	$g = 9.8 \text{ m/s}^2$
f = femto = 10^{-15}	p = pico = 10^{-12}	n = nano = 10^{-9}
μ = micro = 10^{-6}	M = Mega = 10^6	G = Giga = 10^9
Sphere: $A = 4\pi R^2$	$V = \frac{4\pi}{3} R^3$	

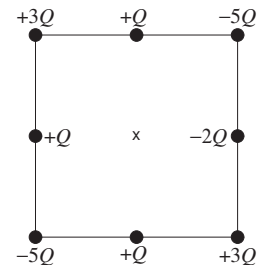
1. Two identical small spheres, each with mass m and charge $6.0 \mu\text{C}$, hang from very light strings, both of length 75 cm (see figure). At equilibrium, the angle $\theta = 30^\circ$. What is m , the mass of each sphere?

- (1) 820 g
(2) 500 g
(3) 450 g
(4) 190 g
(5) 210 g



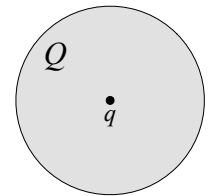
2. Charges are arranged on a square of side d as shown in the diagram. What is the potential at the center of the square (assuming $V = 0$ at infinity)?

- (1) $-3.66kQ/d$
(2) $-1.83kQ/d$
(3) $+7.66kQ/d$
(4) $-7.66kQ/d$
(5) $-7.31kQ/d$



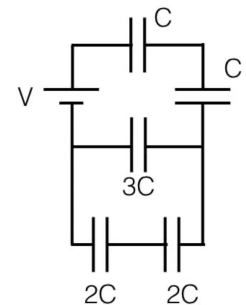
3. The figure shows a nonconducting, uniformly charged sphere of charge Q and radius R with a point charge $q = -\frac{1}{2}Q$ at its center. What is the magnitude of the electric field at the radius $r = R/2$?

- (1) $\frac{3kQ}{2R^2}$ (2) $\frac{5kQ}{8R^2}$ (3) $\frac{kQ}{2R^2}$ (4) $\frac{3kQ}{4R^2}$ (5) $\frac{2kQ}{R^2}$



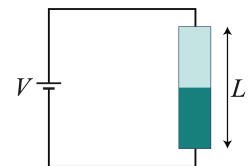
4. The figure shows an arrangement of five capacitors connected to a battery with a potential difference of V . What is the charge on the first (top) C capacitor?

- (1) $\frac{4}{9}CV$
(2) $\frac{1}{3}CV$
(3) $\frac{1}{9}CV$
(4) $\frac{5}{9}CV$
(5) $\frac{2}{3}CV$



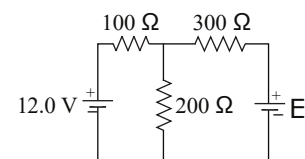
5. A battery is connected to a resistive rod, as shown in the figure. The rod consists of two sections with the same length $L/2$ and cross-sectional area A but different resistivities. The electric field is E in the first half and $2E$ in the second half of the rod, and the current through the rod is $2i$. What is the resistance of the rod?

- (1) $\frac{3}{4}EL/i$ (2) $\frac{3}{2}EL/i$ (3) $\frac{1}{2}EL/i$ (4) $\frac{1}{3}EL/i$ (5) $2EL/i$

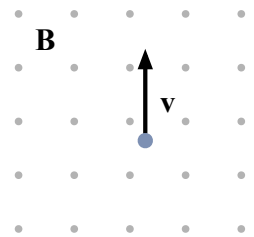


6. In the circuit shown, find the emf E that makes the current through the 100Ω resistor zero.

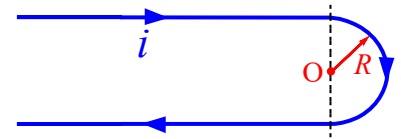
- (1) 30 V (2) 20 V (3) 6.0 V (4) 8.0 V (5) 12 V



7. A helium nucleus (2 protons, 2 neutrons) moves in the direction shown in a magnetic field $\mathbf{B}$ pointing out of the page, as shown in the figure. In what direction is the force experienced by the nucleus?
- (1) right
 - (2) out of page
 - (3) left
 - (4) into the page
 - (5) force is zero



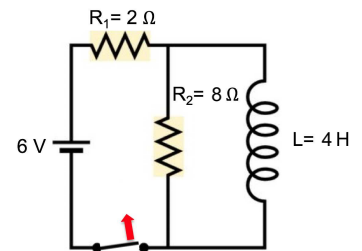
8. An infinite current-carrying wire is turned 180° , as shown in the figure. The radius at which the wire is turned is 1 cm. For what current i is the magnetic field at point O equal to $120 \mu\text{T}$?



- (1) 2.33 A (2) 22 A (3) 14 A (4) 28 A (5) 19 A

9. Consider the circuit shown in the figure. What is the voltage across resistor R_2 a long time after the switch is closed?

- (1) 0 V
- (2) 12 V
- (3) 1.2 V
- (4) 4.8 V
- (5) 6.0 V



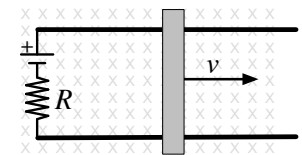
10. A lithium nucleus (3 protons, 4 neutrons) is accelerated through a potential difference of 100,000 V over a distance of 20 cm before entering a uniform magnetic field of magnitude 0.5 T pointing perpendicular to the path of the nucleus. What is the radius of the path followed by the nucleus?

- (1) 14 cm (2) 24 cm (3) 9.1 cm (4) 5.3 cm (5) 7.5 cm

11. A parallel plate capacitor has circular plates of radius $R = 3 \text{ cm}$ and a time-dependent electric field between them of $dE/dt = at$ where $a = 2.5 \times 10^7 \text{ V/m/s}$. What is the magnitude of the induced magnetic field at a radius $r = 4 \text{ cm}$ from the center? Note $r > R$.

- (1) $3.1 \times 10^{-12} \text{ T}$ (2) $4.2 \times 10^{-12} \text{ T}$ (3) $1.1 \times 10^{-9} \text{ T}$ (4) $7.9 \times 10^{-13} \text{ T}$ (5) 0

12. In the figure, a conducting metal bar of mass 0.5 kg sliding along conducting rails is connected to a battery of emf 18 V and resistor $R = 200 \Omega$ to form a circuit. The rails are 30 cm apart and the entire arrangement is embedded in a uniform magnetic field 0.25 T directed into the page, as shown. Find the maximum velocity v the bar attains under these conditions.



- (1) 240 m/s (2) 345 m/s (3) 68 m/s (4) 98 m/s (5) 135 m/s

13. A sinusoidal electromagnetic wave with electric field amplitude of 100 V/m is incident normally on a $2 \text{ cm} \times 3 \text{ cm}$ flat surface and is 60% absorbed. The total energy absorbed in 10 seconds is

- (1) 48 mJ (2) 80 mJ (3) 32 mJ (4) 24 mJ (5) 40 mJ

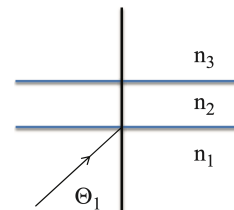
14. A thin converging lens has focal length 8 cm. An object of height 3 cm is placed 12 cm away from the lens. What is the height and orientation of the resulting image?

- (1) 6 cm inverted (2) 6 cm noninverted (3) 9 cm inverted (4) 9 cm noninverted (5) 3 cm inverted

15. A dental hygienist uses a concave mirror to see the back of her patient's teeth. When the mirror is 2.5 cm from a tooth, the image is upright and 3.5 times as large as the tooth. What is the radius of curvature of the mirror?

(1) 7.0 cm (2) 3.7 cm (3) 2.0 cm (4) 3.5 cm (5) 4.0 cm

16. A light ray traveling in a material with index of refraction of n_1 is incident onto a stack of two other materials with indices of refraction n_2 and n_3 , respectively, as shown in the diagram. The light ray has an angle of incidence of $\theta_1 = 45^\circ$ in material n_1 . What angle does the light ray make relative to the normal when it is inside material n_3 ? ($n_1 = 1.1$, $n_2 = 1.3$, $n_3 = 1.4$).



(1) 34° (2) 30° (3) 37° (4) 84° (5) 73°

17. A small gold ring is placed at the bottom of a swimming pool at a depth of 3.0 m. In an attempt to hide the ring, an opaque wooden raft of size $L \times L$ is made to float directly above the ring with the ring just below the center of the raft. What is the *minimum* value of L for which the ring would not be visible for *any* observer above the water ($n = 1.33$)?

(1) 6.8 m (2) 3.4 m (3) 4.5 m (4) 2.9 m (5) 5.2 m

18. Costume jewelry beads are made of glass with index of refraction 1.50. To make them more reflective, they are coated with a layer of material with index of refraction 1.75. What is the approximate minimum coating thickness needed for light of wavelength 550 nm and perpendicular incidence to be reflected with maximum constructive interference?

(1) 79 nm (2) 160 nm (3) 92 nm (4) 180 nm (5) 140 nm

19. The emission spectrum of deuterium (an isotope of hydrogen) is identical to that of hydrogen except that each wavelength λ is slightly decreased by an amount 0.000272λ . Approximately what is the minimum number of lines needed in a diffraction grating that will resolve the deuterium 656 nm emission line from its hydrogen counterpart in the third-order spectrum?

(1) 1230 (2) 3680 (3) 1840 (4) 2240 (5) 4480

20. A single slit diffracts laser light of 612 nm onto a screen 3.0 m away. The distance between the third order minima (located on opposite sides of the central peak) is 8.0 mm. How wide is the slit?

(1) 1.4 mm (2) 0.92 mm (3) 0.69 mm (4) 1.1 mm (5) 0.46 mm